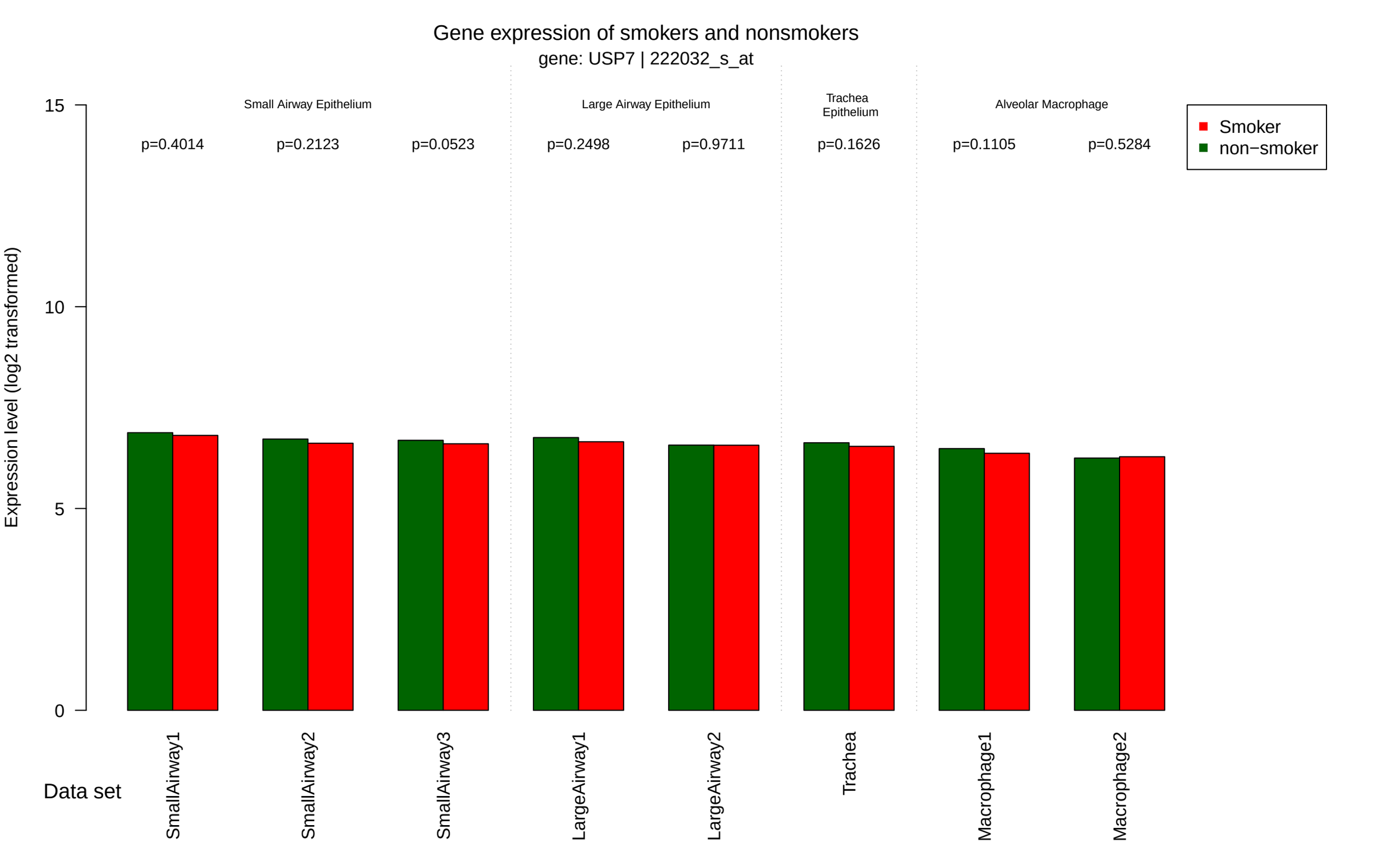
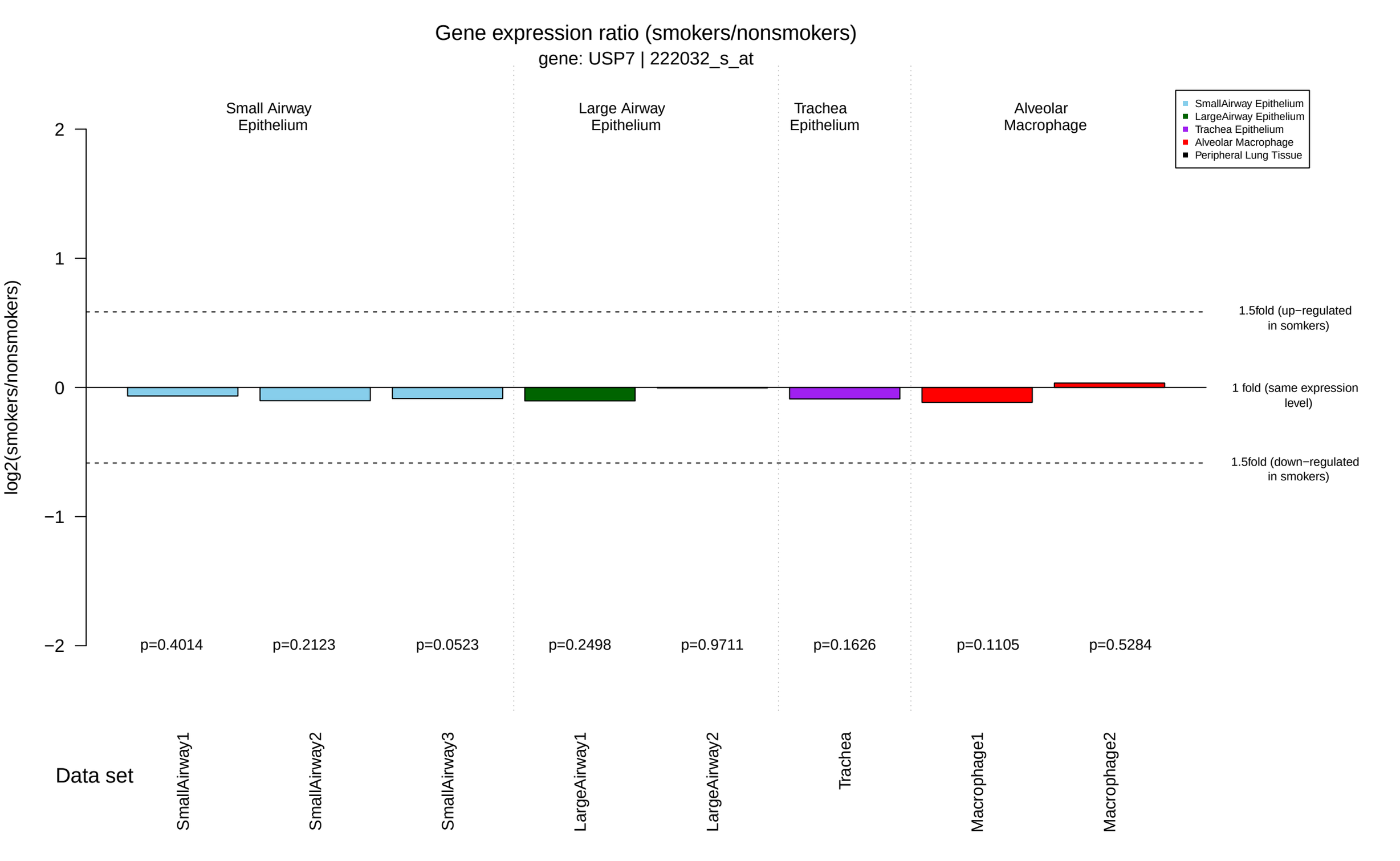
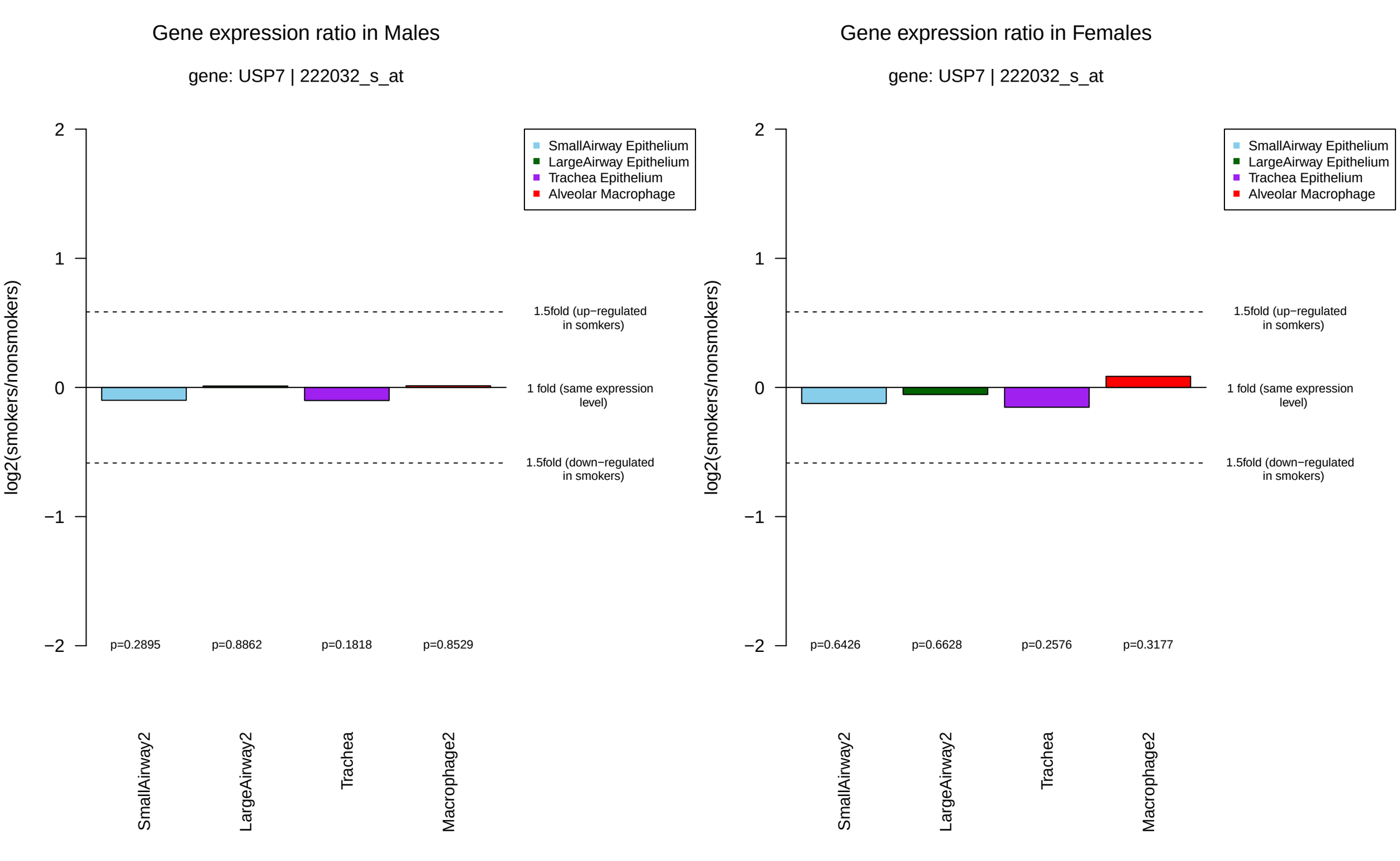
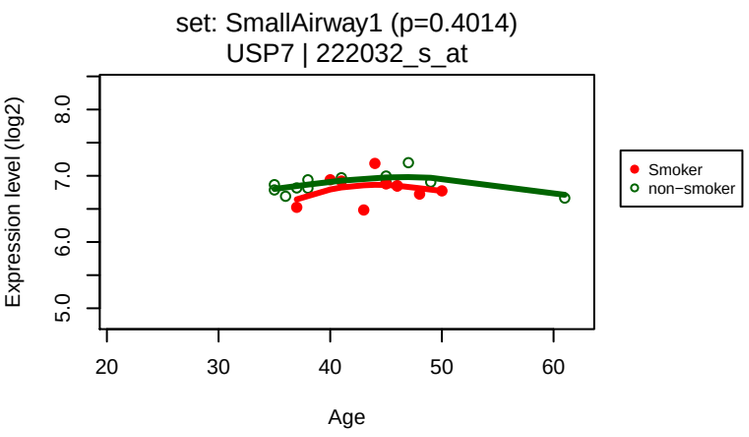
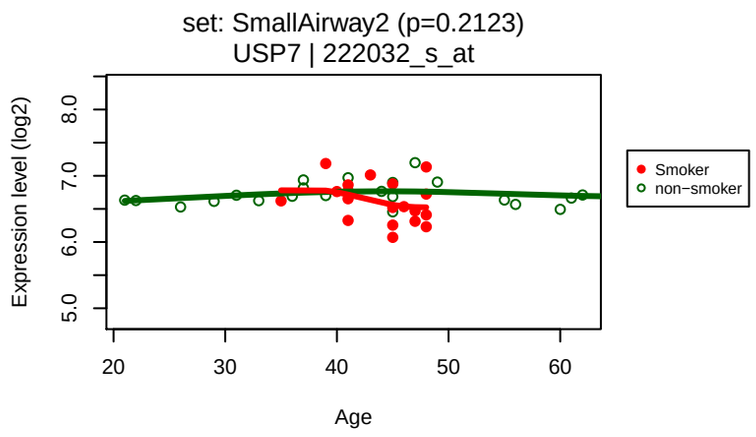
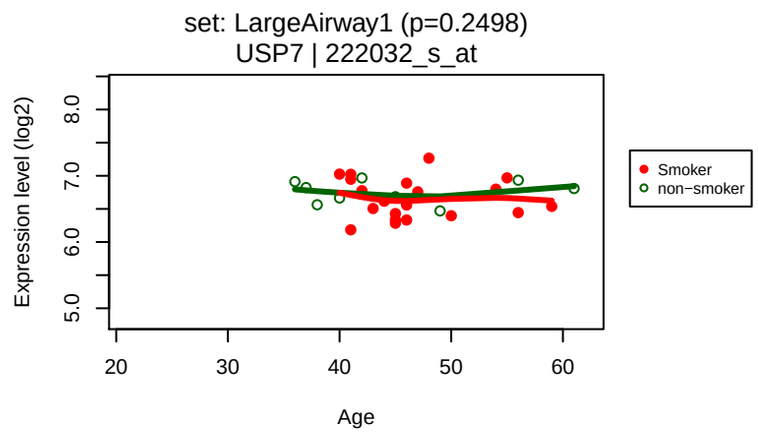

Gene expression of smokers and nonsmokers
gene: USP7 | 222032_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4014
p=0.2123
p=0.0523
p=0.2498
p=0.9711
p=0.1626
p=0.1105
p=0.5284
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

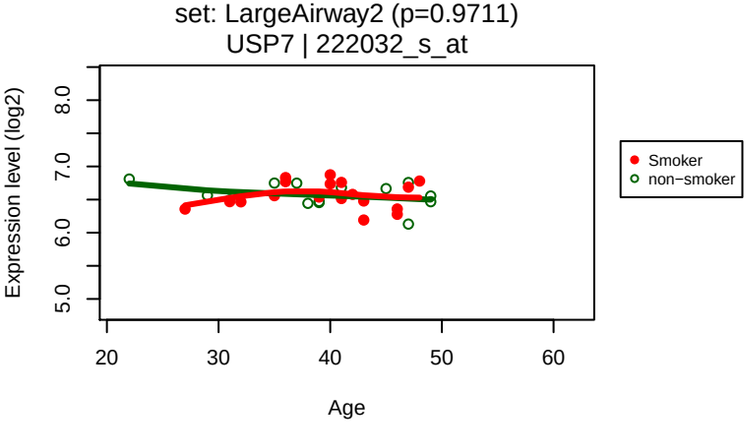

Gene expression ratio (smokers/nonsmokers)
gene: USP7 | 222032_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4014
p=0.2123
p=0.0523
p=0.2498
p=0.9711
p=0.1626
p=0.1105
p=0.5284
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

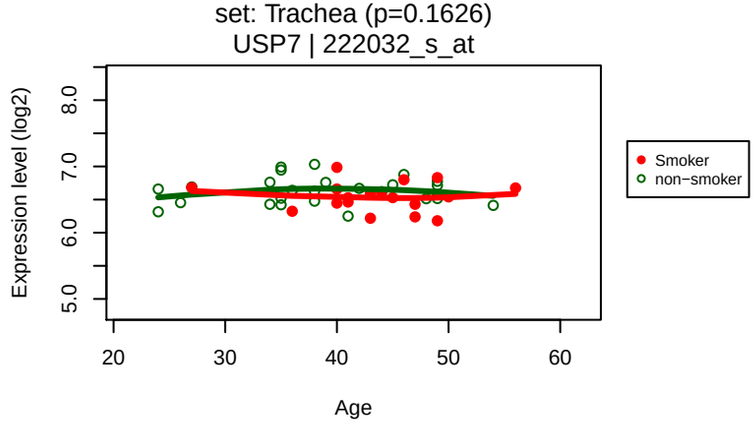

Gene expression ratio in Males
gene: USP7 | 222032_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2895
p=0.8862
p=0.1818
p=0.8529
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP7 | 222032_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6426
p=0.6628
p=0.2576
p=0.3177
SmallAirway2
LargeAirway2
Trachea
Macrophage2

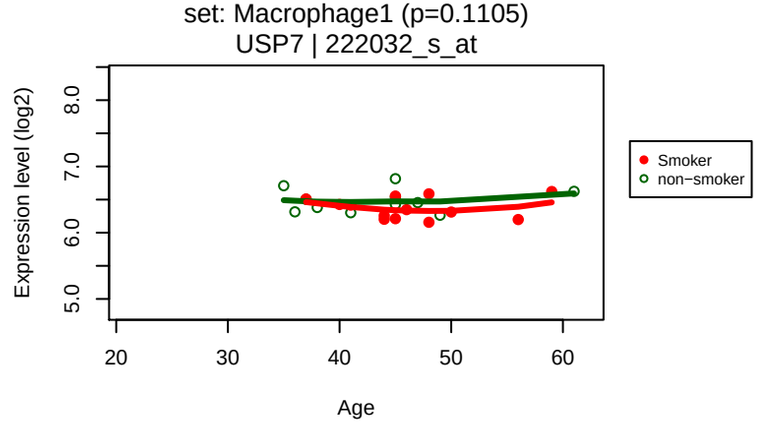

set: SmallAirway1 (p=0.4014)
USP7 | 222032_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker

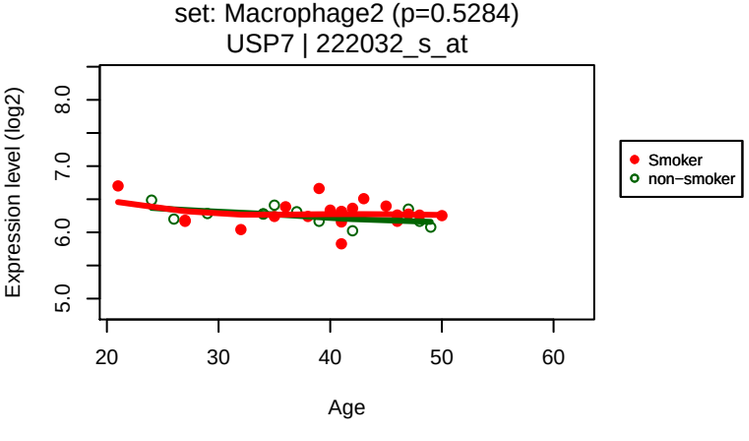

set: SmallAirway2 (p=0.2123)
USP7 | 222032_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker

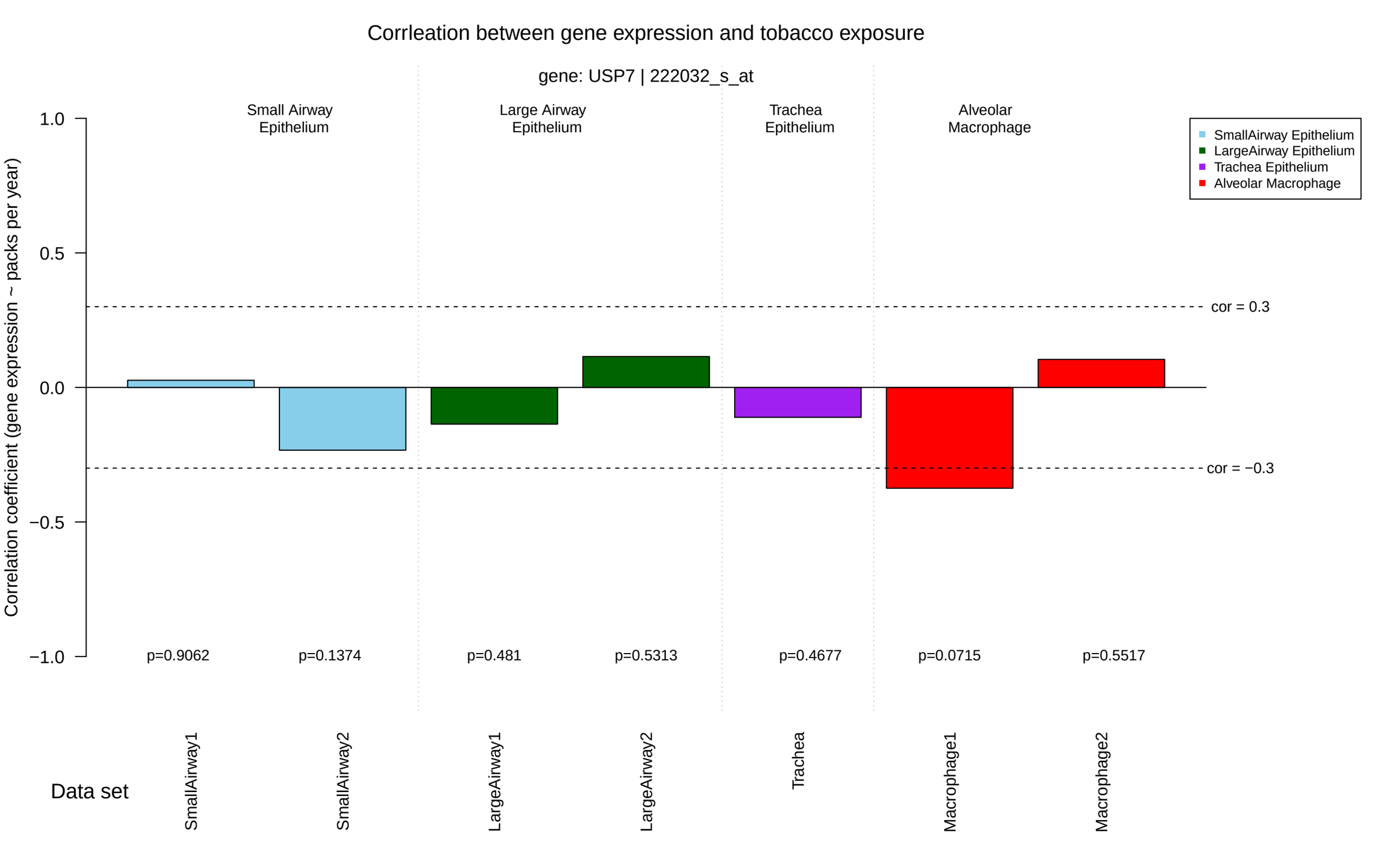

set: LargeAirway1 (p=0.2498)
USP7 | 222032_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9711)
USP7 | 222032_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.1626)
USP7 | 222032_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1105)
USP7 | 222032_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5284)
USP7 | 222032_s_at
Expression level (log2)
5.0
6.0
7.0
8.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP7 | 222032_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9062
p=0.1374
p=0.481
p=0.5313
p=0.4677
p=0.0715
p=0.5517
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2